Supplementary Information

A Mechanistic Model for Radiation-Induced Segregation Controlled Grain-Boundary Oxidation in Austenitic Stainless Steels

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S1 Mesh Sensitivity of the Segregation Profiles and the Finite-Efficiency Sink Condition

Figure S1 shows the Cr, Ni and Si profiles across the grain boundary calculated at four grid spacings Δx with the two boundary conditions for the point defects. With the perfect-sink (Dirichlet) condition the solution does not converge under mesh refinement. As Δx is reduced, the point-defect profiles steepen at the boundary, and both the gradient and the magnitude of the solute segregation increase monotonically with refinement, Fig. S1(a–c). The width of the segregation profiles likewise fails to converge. With the finite-efficiency (Robin) condition, Eqs. (15) and (16) of the main text, the profiles converge under refinement, Fig. S1(d–f).

S2 Partition of the Unoxidized Metal at the Oxidation Front

S2.1 Motivation

If all the unoxidized metal swept by the advancing front were rejected to the metal side of the interface, $\lambda_i = 0$ in Eq. (36) of the main text, the

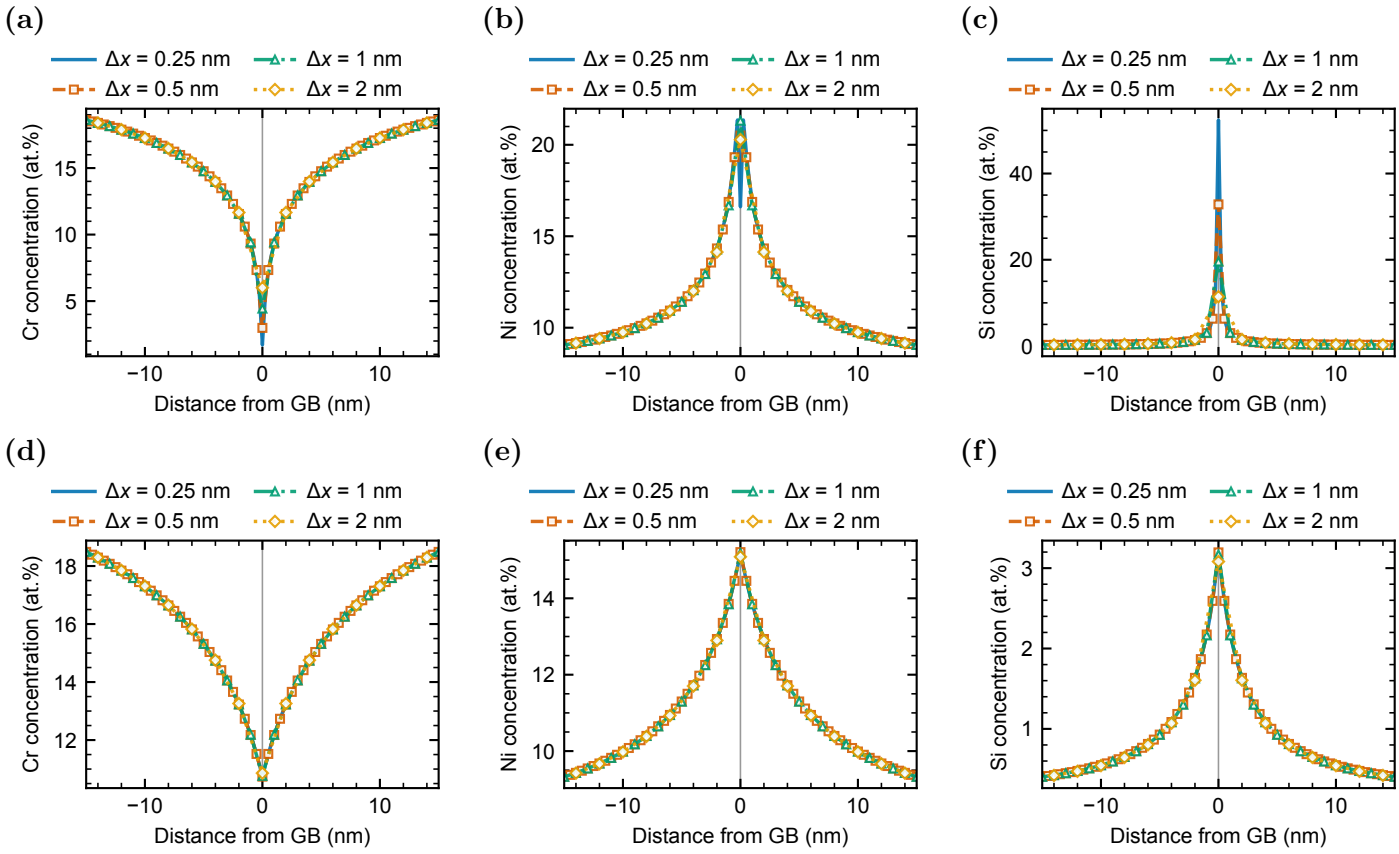


Figure S1: Mesh sensitivity of the calculated Cr, Ni and Si concentration profiles across the grain boundary for the two boundary conditions. (a–c) Dirichlet (perfect-sink) condition, $C_\mathrm{V} = C_\mathrm{V}^\mathrm{eq}$ and $C_\mathrm{I} = C_\mathrm{I}^\mathrm{eq}$ at the boundary, at grid spacings $\Delta x = 0.25$, 0.5, 1 and 2 nm. (d–f) Robin (finite-efficiency) condition, Eqs. (15) and (16) of the main text, at the same spacings.

metal removed at the front would be exactly the metal consumed by the three interfacial reactions. Because the interfacial rate constant of Cr is much larger than that of Fe, this limit depletes Cr and enriches Fe strongly in the metal at the interface. The EDX line scans of Deng et al. [1] show neither (Fig. 4 of the main text). In the irradiated specimens, moreover, the Cr profile has a valley at the centre of the oxide that deepens with dose, which shows that the interfacial Cr concentration follows the radiation-induced segregation profile as the front advances and is not modulated by the

selective oxidation. The excess Fe and Ni must therefore leave the interface by a path other than rejection into the metal.

S2.2 Limiting behaviour

The recession rate of the interface, v_{ox} of Eq. (38) of the main text, has two limits.

- $v_i^{\mathrm{r}} \to \infty$ for every element: $\lambda_i = 0$, the excess is rejected entirely into the metal and $v_{\mathrm{ox}} = \sum_i J_{\mathrm{r}}^i$; the metal removed is the metal oxidized.
- $v_i^{\mathrm{r}} \ll v_{\mathrm{lat}}^{\mathrm{ox,max}}$ for every element: $\lambda_i \to 1$ and the whole swept volume is removed. For every element with $E_i > 0$, $\tilde{J}_{\mathrm{r}}^i = v_{\mathrm{lat}}^{\mathrm{ox,max}} C_i^{\mathrm{GB}}$, and the same holds for the element that sets $v_{\mathrm{lat}}^{\mathrm{ox,max}}$, for which $E_i = 0$ and $J_{\mathrm{r}}^i = v_{\mathrm{lat}}^{\mathrm{ox,max}} C_i^{\mathrm{GB}}$. Since $\sum_i C_i^{\mathrm{GB}} = 1$,

$$v_{\mathrm{ox}} = \sum_i v_{\mathrm{lat}}^{\mathrm{ox,max}} C_i^{\mathrm{GB}} = v_{\mathrm{lat}}^{\mathrm{ox,max}} : \tag{S1}$$

  the interface recedes at the sweep velocity, and the composition profile in the metal is translated rigidly towards the interface without modification by the selective oxidation.

Intermediate values give $\sum_i J_{\mathrm{r}}^i \le v_{\mathrm{ox}} \le v_{\mathrm{lat}}^{\mathrm{ox,max}}$, with the partition set element by element.

S2.3 Relation to solute trapping

Equation (36) of the main text takes its form from the continuous-growth partition coefficient $k(v)$ of solute trapping quoted there. In the solute-trapping theory of rapid solidification [2, 3] a solute atom swept by the interface escapes into the parent phase only by a diffusive exchange jump of length a with interface diffusivity D_i, so that the diffusive velocity is $v_D = D_i/a$. The partition coefficient follows from the competition between the advance of the interface by one monolayer, in a time a/v, and the escape, in a time a^2/D_i; it goes from the equilibrium value k_e for $v \ll v_D$ to complete trapping, $k = 1$, for $v \gg v_D$. Aziz and Kaplan [3] note that the model is not specific to solidification and applies to solid-state transformations. Yu et al. [4] and Sherman et al. [5] transposed it to the oxide/metal interface as nonequilibrium solute capture: oxides form with compositions outside their equilibrium solubility

limits when (i) the captured oxide has a lower free energy than the alloy it forms from and (ii) the effective velocity of cation incorporation exceeds the diffusive velocity of cation exchange across the interface. They argue that for the oxidation of Ni–Cr alloys condition (i) is met over essentially the whole composition range, so that the oxide composition is set by the kinetics, and that condition (ii) holds for most high-temperature and almost all aqueous oxidation. The interface exchange diffusivities that would fix v_D are not known for the Fe–Cr–Ni oxide/metal interface [4], so the rejection velocities v_i^{r} are treated as element-specific parameters; the element that sets $v_{\mathrm{lat}}^{\mathrm{ox,max}}$ (Cr under the present conditions) has $E_i = 0$ and is unaffected by its value.

S3 Cross-Section-Averaged Oxygen Transport and the Lateral Closure

This section derives the one-dimensional oxygen balance, Eqs. (21) and (22) of the main text, and the closure for the interfacial flux, Eqs. (23) and (24) of the main text, from the two-dimensional diffusion equation in the oxide cross-section, and gives the solution procedure for the eigenvalue μ, the neglected higher modes and the range of validity.

S3.1 Notation

The notation is that of the main text: y runs along the grain boundary from the surface, x is normal to it with $x = 0$ on the mid-plane, and only the half-domain $0 \le x \le L(y,t)$ is considered, with L the half-width of Eq. (17) of the main text. $D_{\mathrm{O}}^{\mathrm{eff},y}$ and $D_{\mathrm{O}}^{\mathrm{eff},x}$ are the effective diffusivities along and across the boundary, Eq. (19) of the main text. $C_{\mathrm{O}}(x,y,t)$ denotes the local concentration of dissolved oxygen in the section; its cross-section average and its value at the interface are

$$\bar{C}_{\mathrm{O}}(y,t) = \frac{1}{L}\int_0^L C_{\mathrm{O}}\,\mathrm{d}x, \qquad C_{\mathrm{O}}^{\mathrm{int}} = C_{\mathrm{O}}(L,y,t), \tag{S2}$$

and the diffusive oxygen flux reaching the interface, per unit area of interface and positive towards the metal, is

$$J_{\mathrm{r}}(y,t) = -D_{\mathrm{O}}^{\mathrm{eff},x}\left.\frac{\partial C_{\mathrm{O}}}{\partial x}\right|_{x=L}. \tag{S3}$$

The interfacial reaction law is the total consumption $J_\mathrm{r}(C_\mathrm{O}^\mathrm{int}) = \sum_k J_\mathrm{r}^k(C_\mathrm{O}^\mathrm{int})$ of the rate laws, Eqs. (25)–(27) of the main text; at fixed metal concentrations it is a nondecreasing function of $C_\mathrm{O}^\mathrm{int}$.

S3.2 Assumptions

A1. Equivalent homogeneous medium. The oxide, whose diffusivity $D_\mathrm{O}(x)$ varies across the section, is replaced by a homogeneous anisotropic medium with diffusivity $D_\mathrm{O}^{\mathrm{eff},y}$ along the boundary and $D_\mathrm{O}^{\mathrm{eff},x}$ across it. The two means carry the same steady flux as the actual profile for a uniform gradient along the boundary and for a one-dimensional flux across the section, respectively.

A2. Symmetry at the mid-plane: $\partial C_\mathrm{O}/\partial x = 0$ at $x = 0$.

A3. Slow variation along the boundary: $|\partial L/\partial y| \ll 1$, and L, $D_\mathrm{O}^{\mathrm{eff},x}$, $D_\mathrm{O}^{\mathrm{eff},y}$ and the metal concentrations that enter the reaction law change little over the decay length of the lateral mode, $(L/\mu)\sqrt{D_\mathrm{O}^{\mathrm{eff},y}/D_\mathrm{O}^{\mathrm{eff},x}}$ (Sec. S3.4).

A4. Lateral quasi-steady state: the profile across the section adjusts on the time scale $L^2/D_\mathrm{O}^{\mathrm{eff},x}$, which is short compared with the time scales on which $\bar{C}_\mathrm{O}$ and L change.

A5. Locally, the profile across the section is the fundamental separable mode of the steady diffusion equation (Sec. S3.4). Higher modes decay over a shorter distance along the boundary and are neglected (Sec. S3.6).

No assumption is made on whether the interface is reaction- or diffusion-limited, and C_O is not assumed uniform across the section.

S3.3 Cross-section integration of the two-dimensional balance

Under A1 the two-dimensional conservation equation in the oxide is

$$\frac{\partial C_\mathrm{O}}{\partial t} = \frac{\partial}{\partial y}\left(D_\mathrm{O}^{\mathrm{eff},y}\,\frac{\partial C_\mathrm{O}}{\partial y}\right) + D_\mathrm{O}^{\mathrm{eff},x}\,\frac{\partial^2 C_\mathrm{O}}{\partial x^2}. \tag{S4}$$

Each term is integrated over $0 \le x \le L(y,t)$, with the Leibniz rule where the limit depends on y or t.

Lateral term. With A2 and the definition (S3),

$$\int_0^L D_{\mathrm{O}}^{\mathrm{eff},x}\,\frac{\partial^2 C_{\mathrm{O}}}{\partial x^2}\,\mathrm{d}x = D_{\mathrm{O}}^{\mathrm{eff},x}\,\frac{\partial C_{\mathrm{O}}}{\partial x}\bigg|_0^L = -J_{\mathrm{r}}. \tag{S5}$$

Time term. Because the upper limit moves,

$$\int_0^L \frac{\partial C_{\mathrm{O}}}{\partial t}\,\mathrm{d}x = \frac{\partial}{\partial t}\big(L\,\bar{C}_{\mathrm{O}}\big) - C_{\mathrm{O}}^{\mathrm{int}}\,\frac{\partial L}{\partial t}. \tag{S6}$$

The last term is the dissolved oxygen, at concentration $C_{\mathrm{O}}^{\mathrm{int}}$, contained in the oxide formed during $\mathrm{d}t$.

Along-boundary term. Applying the Leibniz rule twice,

$$\int_0^L \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff},y}\frac{\partial C_{\mathrm{O}}}{\partial y}\right)\mathrm{d}x = \frac{\partial}{\partial y}\int_0^L D_{\mathrm{O}}^{\mathrm{eff},y}\frac{\partial C_{\mathrm{O}}}{\partial y}\,\mathrm{d}x - D_{\mathrm{O}}^{\mathrm{eff},y}\,\frac{\partial C_{\mathrm{O}}}{\partial y}\bigg|_{x=L}\frac{\partial L}{\partial y}, \tag{S7}$$

$$\begin{aligned}\int_0^L D_{\mathrm{O}}^{\mathrm{eff},y}\frac{\partial C_{\mathrm{O}}}{\partial y}\,\mathrm{d}x &= D_{\mathrm{O}}^{\mathrm{eff},y}\left[\frac{\partial}{\partial y}\big(L\,\bar{C}_{\mathrm{O}}\big) - C_{\mathrm{O}}^{\mathrm{int}}\,\frac{\partial L}{\partial y}\right]\\ &= D_{\mathrm{O}}^{\mathrm{eff},y}\,L\,\frac{\partial \bar{C}_{\mathrm{O}}}{\partial y} + D_{\mathrm{O}}^{\mathrm{eff},y}\,(\bar{C}_{\mathrm{O}} - C_{\mathrm{O}}^{\mathrm{int}})\,\frac{\partial L}{\partial y}.\end{aligned} \tag{S8}$$

Hence

$$\begin{aligned}\int_0^L \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff},y}\frac{\partial C_{\mathrm{O}}}{\partial y}\right)\mathrm{d}x = \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff},y}\,L\,\frac{\partial \bar{C}_{\mathrm{O}}}{\partial y}\right)\\ + \underbrace{\frac{\partial}{\partial y}\left[D_{\mathrm{O}}^{\mathrm{eff},y}(\bar{C}_{\mathrm{O}} - C_{\mathrm{O}}^{\mathrm{int}})\frac{\partial L}{\partial y}\right] - D_{\mathrm{O}}^{\mathrm{eff},y}\frac{\partial C_{\mathrm{O}}}{\partial y}\bigg|_{x=L}\frac{\partial L}{\partial y}}_{O(\partial L/\partial y)}.\end{aligned} \tag{S9}$$

Result. Collecting (S5)–(S9),

$$\frac{\partial}{\partial t}\big(L\,\bar{C}_{\mathrm{O}}\big) = \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff},y}\,L\,\frac{\partial \bar{C}_{\mathrm{O}}}{\partial y}\right) - J_{\mathrm{r}} + C_{\mathrm{O}}^{\mathrm{int}}\,\frac{\partial L}{\partial t} + O\left(\frac{\partial L}{\partial y}\right), \tag{S10}$$

which is exact for Eq. (S4). Under A3 the last group is dropped.

At the moving interface the diffusive flux J_r of (S3) supplies both the reactions and the dissolved oxygen in the newly formed oxide (Stefan condition, dissolved oxygen in the metal neglected):

$$J_\mathrm{r} = \sum_k J_\mathrm{r}^k(C_\mathrm{O}^\mathrm{int}) + C_\mathrm{O}^\mathrm{int}\,\frac{\partial L}{\partial t}. \tag{S11}$$

Substituting (S11) into (S10), the terms $C_\mathrm{O}^\mathrm{int}\,\partial L/\partial t$ cancel and the sink is the reaction consumption alone,

$$\frac{\partial}{\partial t}\big(L\,\bar{C}_\mathrm{O}\big) = \frac{\partial}{\partial y}\left(D_\mathrm{O}^{\mathrm{eff},y}\,L\,\frac{\partial \bar{C}_\mathrm{O}}{\partial y}\right) - \sum_k J_\mathrm{r}^k, \tag{S12}$$

which is Eq. (21) of the main text with the flux of Eq. (22). The ratio of the Stefan term in (S11) to the reaction term is the ratio of the dissolved oxygen concentration $C_\mathrm{O}^\mathrm{int}$ to the concentration of the oxygen bound in the oxide, which is much smaller than unity. The diffusive flux J_r and the reaction consumption $\sum_k J_\mathrm{r}^k$ therefore differ by a negligible amount, and the single symbol J_r is used for both, as in the main text; in particular the closure of Sec. S3.5 equates J_r of (S3) to $\sum_k J_\mathrm{r}^k$.

S3.4 Lateral problem: separation of variables

S3.4.1 Local problem

Under A3 and A4, in the neighbourhood of a given position along the boundary the oxide occupies a half-strip $0 \le x \le L$ with L, $D_\mathrm{O}^{\mathrm{eff},x}$, $D_\mathrm{O}^{\mathrm{eff},y}$ and the reaction law treated as constants, and the concentration satisfies the steady form of Eq. (S4),

$$\begin{gathered} D_\mathrm{O}^{\mathrm{eff},x}\,\frac{\partial^2 C_\mathrm{O}}{\partial x^2} + D_\mathrm{O}^{\mathrm{eff},y}\,\frac{\partial^2 C_\mathrm{O}}{\partial y^2} = 0, \\ \left.\frac{\partial C_\mathrm{O}}{\partial x}\right|_{x=0} = 0, \qquad -D_\mathrm{O}^{\mathrm{eff},x}\,\left.\frac{\partial C_\mathrm{O}}{\partial x}\right|_{x=L} = J_\mathrm{r}\big(C_\mathrm{O}(L,y)\big). \end{gathered} \tag{S13}$$

S3.4.2 Separable solutions

With $C_\mathrm{O} = X(x)\,Y(y)$, Eq. (S13) separates into

$$\frac{X''}{X} = -\frac{D_\mathrm{O}^{\mathrm{eff},y}}{D_\mathrm{O}^{\mathrm{eff},x}}\,\frac{Y''}{Y} = -\left(\frac{\mu}{L}\right)^2, \tag{S14}$$

where the separation constant is written as $-(\mu/L)^2$ with $\mu > 0$, because the x-problem must be oscillatory (it has a homogeneous Neumann condition at $x = 0$ and a sink at $x = L$) and the y-dependence must decay into the boundary. Then

$$X = \cos\Big(\frac{\mu x}{L}\Big), \qquad Y = \exp\left(-\frac{\mu y}{L}\sqrt{\frac{D_{\mathrm{O}}^{\mathrm{eff},x}}{D_{\mathrm{O}}^{\mathrm{eff},y}}}\right), \tag{S15}$$

and

$$C_{\mathrm{O}}(x,y) = A\,\cos\Big(\frac{\mu x}{L}\Big)\exp\left(-\frac{\mu y}{L}\sqrt{\frac{D_{\mathrm{O}}^{\mathrm{eff},x}}{D_{\mathrm{O}}^{\mathrm{eff},y}}}\right), \tag{S16}$$

where A is an amplitude and μ is to be fixed by the interfacial condition. The dependence on x involves $D_{\mathrm{O}}^{\mathrm{eff},x}$ alone; the ratio of the two diffusivities enters only the decay length along the boundary, $(L/\mu)\sqrt{D_{\mathrm{O}}^{\mathrm{eff},y}/D_{\mathrm{O}}^{\mathrm{eff},x}}$. Different values of μ give different modes; the fundamental mode has the smallest μ and the slowest decay in y, and by A5 it is the one retained.

S3.4.3 Averages and flux of the mode

From (S16), with the factor depending on y absorbed into A,

$$\bar{C}_{\mathrm{O}} = \frac{A}{L}\int_0^L \cos\Big(\frac{\mu x}{L}\Big)\,\mathrm{d}x = A\,\frac{\sin\mu}{\mu}, \tag{S17}$$

$$C_{\mathrm{O}}^{\mathrm{int}} = A\cos\mu, \tag{S18}$$

$$J_{\mathrm{r}} = -D_{\mathrm{O}}^{\mathrm{eff},x}\,\frac{\partial C_{\mathrm{O}}}{\partial x}\bigg|_{x=L} = \frac{D_{\mathrm{O}}^{\mathrm{eff},x}\,A\,\mu\sin\mu}{L}. \tag{S19}$$

Eliminating the amplitude with (S17), $A = \mu\bar{C}_{\mathrm{O}}/\sin\mu$, gives the two relations of Eq. (23) of the main text, $J_{\mathrm{r}} = D_{\mathrm{O}}^{\mathrm{eff},x}\mu^2\bar{C}_{\mathrm{O}}/L$ and $C_{\mathrm{O}}^{\mathrm{int}} = \bar{C}_{\mathrm{O}}\,\mu\cot\mu$. Since $\mu\cot\mu$ decreases monotonically from 1 at $\mu = 0$ to 0 at $\mu = \pi/2$, $0 \le C_{\mathrm{O}}^{\mathrm{int}} \le \bar{C}_{\mathrm{O}}$ for $\mu \in [0, \pi/2]$.

S3.4.4 Consistency with the averaged equation

Substituting the mode into the steady form of (S12) with constant L, $D_{\mathrm{O}}^{\mathrm{eff},x}$ and $D_{\mathrm{O}}^{\mathrm{eff},y}$,

$$D_{\mathrm{O}}^{\mathrm{eff},y}L\,\frac{\partial^2\bar{C}_{\mathrm{O}}}{\partial y^2} = D_{\mathrm{O}}^{\mathrm{eff},y}L\,\frac{D_{\mathrm{O}}^{\mathrm{eff},x}}{D_{\mathrm{O}}^{\mathrm{eff},y}}\left(\frac{\mu}{L}\right)^2\bar{C}_{\mathrm{O}} = \frac{D_{\mathrm{O}}^{\mathrm{eff},x}\,\mu^2\,\bar{C}_{\mathrm{O}}}{L} = J_{\mathrm{r}}, \tag{S20}$$

with no residual. The averaged equation, which carries $D_{\mathrm{O}}^{\mathrm{eff},y}$, and the mode closure, which carries $D_{\mathrm{O}}^{\mathrm{eff},x}$, are therefore mutually consistent; no additional approximation is introduced by combining them.

S3.5 The equation for μ

S3.5.1 General reaction law

Imposing the interfacial condition of (S13) on the mode gives Eq. (24) of the main text, written here as the root of

$$g(\mu) \equiv \frac{D_{\mathrm{O}}^{\mathrm{eff},x}\,\mu^2\,\bar{C}_{\mathrm{O}}}{L} - J_{\mathrm{r}}\big(\bar{C}_{\mathrm{O}}\,\mu\cot\mu\big) = 0. \tag{S21}$$

It is a single scalar equation in μ; the inputs are the local values of $\bar{C}_{\mathrm{O}}$, L, $D_{\mathrm{O}}^{\mathrm{eff},x}$ and the metal concentrations that enter $J_{\mathrm{r}}(\cdot)$. The diffusivity $D_{\mathrm{O}}^{\mathrm{eff},y}$ does not enter.

Existence and uniqueness. On $(0,\pi/2)$ the first term increases monotonically from 0; the second term decreases monotonically, because $J_{\mathrm{r}}(\cdot)$ is nondecreasing and $\mu\cot\mu$ is decreasing. Hence g is strictly increasing, with

$$g(0^+) = -J_{\mathrm{r}}(\bar{C}_{\mathrm{O}}), \qquad g(\pi/2) = \frac{\pi^2 D_{\mathrm{O}}^{\mathrm{eff},x}\,\bar{C}_{\mathrm{O}}}{4L} - J_{\mathrm{r}}(0). \tag{S22}$$

If $J_{\mathrm{r}}(\bar{C}_{\mathrm{O}}) > 0$ (oxidation possible at the local average concentration) and $J_{\mathrm{r}}(0) \le 0$ (no consumption at zero oxygen, which holds for all three rate laws), there is exactly one root $\mu \in (0,\pi/2)$. If $J_{\mathrm{r}}(\bar{C}_{\mathrm{O}}) \le 0$, no oxygen is consumed: $\mu = 0$, $J_{\mathrm{r}} = 0$ and $C_{\mathrm{O}}^{\mathrm{int}} = \bar{C}_{\mathrm{O}}$.

Newton iteration. The derivative is

$$\begin{aligned} g'(\mu) &= \frac{2D_{\mathrm{O}}^{\mathrm{eff},x}\,\mu\,\bar{C}_{\mathrm{O}}}{L} - J_{\mathrm{r}}'\big(C_{\mathrm{O}}^{\mathrm{int}}\big)\,\bar{C}_{\mathrm{O}}\,\frac{\mathrm{d}}{\mathrm{d}\mu}\big(\mu\cot\mu\big), \\ \frac{\mathrm{d}}{\mathrm{d}\mu}\big(\mu\cot\mu\big) &= \frac{\sin\mu\cos\mu - \mu}{\sin^2\mu} < 0, \end{aligned} \tag{S23}$$

so $g' > 0$ throughout. A safeguarded Newton iteration is used, which falls back to bisection if an iterate leaves $(0,\pi/2)$, started from the linearized value of Sec. S3.5.2.

Procedure at each grid point and time step.

1. Inputs: $\bar{C}_\mathrm{O}$, L, $D_\mathrm{O}^{\mathrm{eff},x}$, $C_\mathrm{Cr}^\mathrm{GB}$, $C_\mathrm{Fe}^\mathrm{GB}$, $C_\mathrm{Si}^\mathrm{GB}$.
2. If $J_\mathrm{r}(\bar{C}_\mathrm{O}) \le 0$: $\mu = 0$, $J_\mathrm{r} = 0$, $C_\mathrm{O}^\mathrm{int} = \bar{C}_\mathrm{O}$.
3. Otherwise $g(\mu) = 0$ is solved on $(0, \pi/2)$.
4. $J_\mathrm{r} = D_\mathrm{O}^{\mathrm{eff},x}\mu^2\bar{C}_\mathrm{O}/L$ and $C_\mathrm{O}^\mathrm{int} = \bar{C}_\mathrm{O}\,\mu\cot\mu$; the individual $J_\mathrm{r}^k(C_\mathrm{O}^\mathrm{int})$ give the metal sinks, Eqs. (28)–(30) of the main text, and the thickness rates, Eqs. (31)–(33).
5. J_r is the sink in Eq. (21) of the main text.

S3.5.2 Linear reaction law

For $J_\mathrm{r} = k\,C_\mathrm{O}^\mathrm{int}$, with $k = \mathrm{d}J_\mathrm{r}/\mathrm{d}C_\mathrm{O}^\mathrm{int}$ as in the main text, Eq. (S21) becomes

$$\frac{D_\mathrm{O}^{\mathrm{eff},x}\mu^2}{L} = k\,\mu\cot\mu \qquad \Longleftrightarrow \qquad \mu\tan\mu = \frac{kL}{D_\mathrm{O}^{\mathrm{eff},x}} \equiv \mathrm{Da}, \tag{S24}$$

the classical eigenvalue condition for a Robin boundary. Da is the ratio of the diffusion resistance across the section, $L/D_\mathrm{O}^{\mathrm{eff},x}$, to the reaction resistance $1/k$. The concentration $\bar{C}_\mathrm{O}$ drops out: μ depends only on Da (Table S1).

Table S1: Fundamental mode for a linear reaction law.

$\mathrm{Da} = kL/D_\mathrm{O}^{\mathrm{eff},x}$	μ	$C_\mathrm{O}^\mathrm{int}/\bar{C}_\mathrm{O} = \mu\cot\mu$	$J_\mathrm{r}L/(D_\mathrm{O}^{\mathrm{eff},x}\bar{C}_\mathrm{O}) = \mu^2$
0	0	1	0
0.1	0.311	0.968	0.097
1	0.860	0.740	0.740
10	1.429	0.204	2.042
∞	$\pi/2$	0	$\pi^2/4 \approx 2.467$

In the reaction-limited limit, $\mathrm{Da} \ll 1$, the profile across the section is nearly uniform, with a parabolic deviation of relative size $O(\mathrm{Da})$, and the decay length along the boundary is $(L/\mu)\sqrt{D_\mathrm{O}^{\mathrm{eff},y}/D_\mathrm{O}^{\mathrm{eff},x}} \simeq \sqrt{D_\mathrm{O}^{\mathrm{eff},y}L/k}$, much larger than L. In the diffusion-limited limit, $\mathrm{Da} \gg 1$, the profile across the section is a quarter cosine and the decay length along the boundary is $(2L/\pi)\sqrt{D_\mathrm{O}^{\mathrm{eff},y}/D_\mathrm{O}^{\mathrm{eff},x}}$. Between these limits μ interpolates smoothly.

S3.6 Range of validity

1. **Higher modes.** The interfacial condition admits a sequence of eigenvalues $\mu_0 < \mu_1 < \ldots$, with $\mu_n \in (n\pi,\, n\pi + \pi/2)$ for a Robin condition. Each mode decays along the boundary over the distance $(L/\mu_n)\sqrt{D_{\mathrm{O}}^{\mathrm{eff},y}/D_{\mathrm{O}}^{\mathrm{eff},x}}$. The first higher mode therefore decays over at most $(L/\pi)\sqrt{D_{\mathrm{O}}^{\mathrm{eff},y}/D_{\mathrm{O}}^{\mathrm{eff},x}}$, compared with at least $(2L/\pi)\sqrt{D_{\mathrm{O}}^{\mathrm{eff},y}/D_{\mathrm{O}}^{\mathrm{eff},x}}$ for the fundamental. Higher modes are excited where the boundary data change abruptly: the surface entry, the oxidation tip, and any position where L, the diffusivities or the metal concentrations jump. Away from such places only the fundamental survives (A5).

2. **Slowly varying geometry (A3).** The mode is derived for a strip of constant width. For a wedge-shaped oxide the closure is applied with the local $L(y)$; the error is $O(\partial L/\partial y)$, the same order as the terms dropped in (S10).

3. **Quasi-steady lateral profile (A4).** $L^2/D_{\mathrm{O}}^{\mathrm{eff},x}$ must be short compared with the time scales of $\bar{C}_{\mathrm{O}}(t)$ and $L(t)$.

4. **Early stage.** When the depth of the oxidized zone is comparable to the decay length $(L/\mu)\sqrt{D_{\mathrm{O}}^{\mathrm{eff},y}/D_{\mathrm{O}}^{\mathrm{eff},x}}$, the whole problem is two-dimensional. The one-dimensional description is meaningful only once the oxidized zone is slender; the initial stage lies outside its scope.

5. **Equivalent homogeneous medium (A1).** The harmonic mean $D_{\mathrm{O}}^{\mathrm{eff},x}$ is exact for a one-dimensional steady flux across the section; for a nonuniform $D_{\mathrm{O}}(x)$ the fundamental mode is not a pure cosine, and the deviation is neglected. The arithmetic mean $D_{\mathrm{O}}^{\mathrm{eff},y}$ is exact when the gradient along the boundary is uniform across the section; the correlation between $D_{\mathrm{O}}(x)$ and the variation of $\partial C_{\mathrm{O}}/\partial y$ across the section is neglected.

S4 Apparent Effective Diffusivities of the Oxidation Front

Table S2 lists the apparent effective diffusivities $\bar{D}_{\mathrm{eff}} = L^2/2t$ of the two experimental series used in the main text, the 500 h series of Deng et al. (2017) [1] and the 1500 h series of Deng et al. (2020) [6], computed from the reported final penetration depth L and total exposure time t. The values assume parabolic kinetics from $t = 0$ with no incubation correction.

Table S2: Apparent effective diffusivities of the grain-boundary oxidation front, $\bar{D}_{\mathrm{eff}} = L^2/2t$, from the reported final penetration depth L and total exposure time t. Depths for Deng et al. (2017) [1] are digitized from Fig. 12 therein (mean over the reported boundaries); depths for Deng et al. (2020) [6] are the values reported in the text.

Source	Dose	t (h)	L (nm)	$\bar{D}_{\mathrm{eff}}$ (nm^2/h)	$\bar{D}_{\mathrm{eff}}$ (10^{-22} m^2/s)
Deng et al. (2017) [1]	SA	500	40	1.6	4.4
Deng et al. (2017) [1]	0.5 dpa	500	62	3.8	10.7
Deng et al. (2017) [1]	3.0 dpa	500	105	11.0	30.6
Deng et al. (2020) [6]	0.5 dpa	1500	190	12.0	33.4
Deng et al. (2020) [6]	1.5 dpa	1500	240	19.2	53.3

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